Yuyang Wang

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Education

University of Alberta *Edmonton, AB, Canada*
B.Sc. Honors in Computing Science, GPA: 3.96/4.00 *Sep. 2024 – Apr. 2027 (expected)*

- Awards: University of Alberta Undergraduate Scholarship (International), 2025.

Shanghai University *Shanghai, China*
Computer Science and Technology (transferred to University of Alberta), GPA: 3.61/4.00 *Sep. 2022 – May 2024*

- Awards: Academic Progress Scholarship and Innovation & Entrepreneurship Scholarship, Shanghai University, 2024.

Experience

Paid Undergraduate Research Assistant *May 2026 – Present*
University of Alberta — Supervisor: Professor Zhou Yang, Canada CIFAR AI Chair *Edmonton, AB, Canada*

Real-World, Test-Grounded Deprecated API Update: a benchmark for LLM-based migration of breaking third-party Python library API changes. First-author manuscript in preparation for ISSTA 2027; scheduled for presentation at CSER 2026.

- Led the development of a test-grounded benchmark for real-world Python API migration, constructing 1,464 migration tasks from 405 repositories, each with upgrade-induced test failures and test-verified reference fixes.
- Built an API-evolution knowledge base for 59 Python libraries: enumerated the public API over 5,900+ releases, diffed adjacent releases into 1.6M change events over 214K APIs, built static call graphs within each release, and mined 37K real-world repositories from 224M GitHub repositories (The Stack v3) with an AST-based import resolver.
- Developed fault localization combining spectrum-based scoring and differential execution tracing to rank diverging API calls; categorized 90K API-change instances using an 18-category taxonomy and LLM-assisted annotation.
- Designed a repair-agent harness and benchmarked 3 LLMs under 7 information settings on 30 hand-verified tasks: tracebacks plus library-source tools raised Qwen3.8-27B from 0/30 to 14/30 resolved without code execution and to 23/30 with code execution.

Teaching Assistant *Sep. 2025 – Apr. 2026*
University of Alberta, Department of Computing Science *Edmonton, AB, Canada*

- TA for CMPUT 365 (Introduction to Reinforcement Learning) and CMPUT 201 (Practical Programming Methodology): led labs and office hours for 200+ undergraduates in C, Linux and RL; contributed RL exam questions.

Projects

Local 2D SLAM with Dynamic Path Planning *Nov. 2025 – Dec. 2025*
C++, ROS, Ceres Solver, Docker *Code* *Report*

- Developed a local 2D LiDAR SLAM pipeline combining EKF-filtered odometry with Ceres-based scan-to-map optimization (bicubic B-spline interpolation) and 2-cm probabilistic occupancy-grid mapping with Bresenham ray tracing.
- Integrated obstacle-inflated A* planning with gradient-descent path smoothing and Pure Pursuit control in a multi-threaded ROS navigation pipeline; tested in Gazebo/RViz and on a physical Clearpath Jackal robot.

Deep Reinforcement Learning: Value-Based and Policy-Based Methods *Jan. 2026 – Mar. 2026*
Graduate Coursework (CMPUT 628), PyTorch, Gymnasium, MuJoCo *Code*

- Implemented DQN and Double DQN with experience replay, target networks, and multi-step TD returns (1, 5, 10); extended Double DQN to image-based Jumping Tasks using 40-step returns and achieved a 99.7% success rate.
- Developed PPO from scratch with Gaussian actor-critic policies, GAE, and clipped policy optimization (mean return of about 3,200 on Walker2d-v4 and 2,700 on Ant-v4 over 3 seeds); ran ablations on entropy regularization, target update frequency, and replay buffer capacity.

Technical Skills

Programming Languages: Python, C++, Java, SQL, Bash, MATLAB
LLM & ML: PyTorch, Hugging Face Transformers, vLLM, tool-calling agents and MCP servers, Gymnasium, MuJoCo
Software Engineering: Docker, uv, Python AST & runtime tracing, Git, Linux, Slurm, Fuzzing, SMT solver, SQLite
Robotics: ROS, tf2, Gazebo, RViz, Ceres Solver